

CNN-ENHANCED DISCRETE WAVELET TRANSFORM FOR EFFICIENT LOSSLESS MEDICAL IMAGE COMPRESSION

CSI3019-ADVANCED DATA COMPRESSION TECHNIQUES

Under the Supervision of

Dr. Vishnu Priya A

Professor

School of Computer Science and Engineering (SCOPE)

Submitted by

Team no: 08

Haarini SK – 22MID0231 – haarini.sk2022@vitstudent.ac.in - 6381363538

Sivaranjani T.S -22MID0232- sivaranjani.ts2022@vitstudent.ac.in - 8668010516

Drishna B-22MID0273- drishna.b2022@vitstudent.ac.in - 7904062746

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ABSTRACT

In healthcare systems, medical picture compression is essential for effective transmission and storage while preserving diagnostic quality. This study introduces a novel hybrid method for lossless medical picture compression that combines Convolutional Neural Networks (CNNs) with Discrete Wavelet Transform (DWT). The suggested technique improves the DWT coefficient prediction by utilizing CNN's feature learning capabilities, which leads to higher compression ratios without information loss. In contrast to conventional wavelet-based compression methods that just employ mathematical changes, our method makes use of deep learning to more efficiently take advantage of spatial redundancies and correlations in medical images. Better residual encoding is made possible by the CNN architecture's ability to predict high-frequency wavelet coefficients from low-frequency components. Experimental validation on several medical imaging modalities, such as MRI, CT, and X-ray, shows that the suggested CNN-enhanced DWT method maintains correct reconstruction while achieving compression ratios that are 15–23% greater than traditional DWT-based methods. CNN's incorporation of learnt priors greatly lowers encoded coefficient entropy, increasing the efficiency of the ensuing lossless entropy coding. This development provides a computationally effective approach appropriate for real-time clinical applications, addressing the rising problem of medical data management in contemporary healthcare infrastructure.

KEYWORDS: Medical image compression, Discrete Wavelet Transform, Convolutional Neural Networks, Lossless compression, Deep learning, Entropy coding, Healthcare informatics

1.INTRODUCTION

Healthcare data management now faces previously unheard-of difficulties due to the exponential growth of digital medical imaging. High-resolution medical images, such as MRI scans, CT scans, X-rays, ultrasounds, and pathology slides, are produced in enormous quantities every day by contemporary diagnostic methods. Several terabytes of medical imaging data might be generated daily by a normal hospital, placing a burden on network bandwidth and storage facilities during transmission and archival procedures. Effective compression methods that may drastically cut down on transmission time and storage needs while preserving excellent image fidelity for precise clinical diagnosis are required due to the explosion of data. Due to strict regulatory and diagnostic criteria, medical image compression is fundamentally different from general-purpose image compression. Even small data loss in clinical settings might result in incorrect diagnoses, endangering patient safety and treatment results. As a result, in many medical imaging applications where diagnostic accuracy cannot be compromised, lossless compression techniques are required. However, the moderate compression ratios (2:1 to 4:1) that are usually achieved by traditional lossless compression algorithms limit their ability to effectively handle the scope of contemporary medical data concerns. The capacity of Discrete Wavelet Transform

(DWT) to break down images into multi-resolution frequency components has made it a potent tool for compressing medical images. By concentrating important information in low-frequency coefficients, DWT offers superior energy compaction, whereas high-frequency coefficients may contain redundant or less important features. Medical photos, whose anatomical structures show spatial association across several scales, are a good fit for this hierarchical approach. Numerous DWT-based compression strategies have been put forth, combining entropy coding methods (Huffman, Arithmetic, LZW) and wavelet families (Haar, Daubechies, Biorthogonal). Despite their effectiveness, traditional DWT-based techniques have built-in drawbacks. They depend on set mathematical changes that might not be the best way to accommodate the various features of various medical imaging modalities. Complex, non-linear correlations found in medical imaging may be missed when high-frequency coefficients are predicted from low-frequency components using preset mathematical formulas. Furthermore, acquired patterns unique to the anatomy and texture of medical images are not utilized by conventional approaches. Convolutional Neural Networks (CNNs), a recent development in deep learning, have transformed image processing tasks by directly learning hierarchical feature representations from data. CNNs are excellent at identifying intricate patterns and spatial connections that are challenging to represent using manually written methods. Researchers are now investigating their potential in picture compression as a result of their success in image classification, segmentation, and enhancement. Nevertheless, there is still a lack of

research on the combination of deep learning with conventional transform coding for lossless medical picture compression. For effective lossless medical picture compression, this research suggests a novel hybrid strategy that combines CNNs with DWT in a synergistic manner. The main breakthrough is the use of a well-thought-out CNN architecture to forecast high-frequency wavelet coefficients from low-frequency components, producing forecasts that are more accurate and have fewer residuals. Because of their lower entropy, these reduced residuals allow for more effective lossless entropy coding. During training, the CNN picks up modality-specific patterns, adjusting to the distinct features of various forms of medical imaging.

The main contributions of this work are:

1. **Novel Hybrid Architecture:** A CNN-enhanced DWT system that combines conventional wavelet transform with deep learning-based coefficient prediction for increased compression effectiveness.
2. **Adaptive Learning:** CNN-based prediction that learns textural patterns and spatial correlations unique to several medical imaging modalities (MRI, CT, X-ray).
3. **Enhanced Compression Ratios:** While preserving flawless lossless reconstruction, it has been shown that compression ratios can be improved by 15–23% when compared to traditional DWT-based techniques.
4. **Computational Efficiency:** A well-designed architecture with a manageable computational overhead that can be used in real-time clinical deployment.
5. **Thorough Evaluation:** Comparison with cutting-edge lossless compression methods and extensive experimental validation across a variety of medical imaging modalities.

This paper's remaining sections are arranged as follows: Section 2 examines relevant research in deep learning techniques, wavelet-based techniques, and medical picture compression. The suggested CNN-enhanced DWT methodology is described in depth in Section 3. The experimental setup and implementation are described in Section 4. Results and comparative analysis are covered in Section 5. The work is finally concluded with future research directions in Section 6.

2. LITERATURE REVIEW

2.1 WAVELET-BASED MEDICAL IMAGE COMPRESSION

The multi-resolution analytical capabilities of the Discrete Wavelet Transform have led to substantial research into its application in medical picture reduction. DWT-based hybrid adaptive lossless image compression framework [1] showed how adaptive wavelet base and decomposition level selection might enhance compression efficiency. The study focused on entropy analysis of wavelet coefficients and shown that clever adaptation to image properties outperforms fixed transformation strategies.

The computational performance requirement for real-time clinical applications was solved by a VLSI architecture for lossless compression of medical pictures utilizing DWT [2]. The hardware implementation showed that specialized architectures may greatly speed up wavelet-based compression, resulting in real-time processing speeds appropriate for incorporation into medical imaging devices. This study made clear how crucial it is to take implementation limitations into account in addition to compression performance. The goal of the integer wavelet transform for effective compression of medical images [3] was to minimize computational complexity without sacrificing compression efficiency. By doing away with floating-point operations, integer wavelet transformations allow for quicker processing and easier hardware implementation. Integer approximations of continuous wavelets could achieve similar compression ratios with a substantially lower computing overhead, according to experiments conducted on a variety of medical imaging modalities.

By storing differences between successive coefficients rather than absolute values, an enhanced wavelet difference reduction-based technique for compressing medical images [6] took advantage of the redundancy in wavelet coefficients. Better compression ratios were obtained by combining this differential encoding with context-adaptive arithmetic coding, especially for medical images with moderate intensity shifts and smooth areas. The technique showed that compression is improved beyond normal wavelet decomposition by the clever use of inter-coefficient correlation. Effective image compression methods that use quantization and wavelet transform [7] examined the trade-offs between image quality and compression ratio. Adaptive encoding systems that might be expanded to lossless

scenarios through improved coefficient representation were designed using the quantization strategies for wavelet coefficients. In order to maintain important diagnostic information in tumor regions and achieve higher compression in non-pathological areas, an effective compression technique that uses a modified wavelet transform designed for brain tumor pictures [9] modified the wavelet decomposition strategy. This region-of-interest method showed how task-specific wavelet parameter adjustment could improve compression for certain medical imaging uses.

2.2 HYBRID TRANSFORM-BASED APPROACHES

To take advantage of complimentary qualities, a number of researchers have investigated merging different transformation approaches. Image compression for biomedical imaging utilizing DCT and LZW lossless method [4] used Lempel-Ziv-Welch (LZW) dictionary-based encoding after frequency decomposition using Discrete Cosine Transform. This combination enhanced compression ratios for medical images with repeating structures by taking advantage of the energy compaction of transform coding and the pattern recognition capabilities of dictionary coding. A cascaded strategy that applied many transformations consecutively, each stage taking advantage of different sorts of redundancy, was proposed by an effective JPEG picture compression method based on the Haar wavelet transform, discrete cosine transform, and run length encoding approaches [5]. Run-length encoding recorded patterns of spatial repetition, DCT improved frequency analysis, and the Haar wavelet offered coarse multi-resolution breakdown. The multi-stage concept impacted design principles for lossless hybrid approaches, even though it was created for lossy compression in industrial processes.

2.3 COMPARATIVE STUDIES AND BENCHMARKING

Through a thorough examination across modalities (CT, MRI, X-ray, and ultrasound), a comparative analysis of lossless image compression algorithms based on several medical image types [8] showed that compression efficiency varies greatly depending on image features. In contrast to images with smoother areas (like MRI), images with more structural complexity and texture (like ultrasound) have lower compression ratios. This study set benchmarks for assessing novel compression techniques and underlined the significance of modality-specific optimization. Alternative wavelet families (Haar, Daubechies, Symlets, Coiflets, and Biorthogonal) and decomposition levels were thoroughly examined in a comparative assessment of alternative wavelet approaches for image compression [10]. The results showed that no single wavelet basis compresses all image types optimally; instead, the selection is based on the features of the image, including texture complexity, edge density, and frequency distribution. This work provided empirical evidence supporting adaptive wavelet selection strategies.

2.4 DEEP LEARNING IN IMAGE COMPRESSION

Recent developments in deep learning have created new opportunities for picture compression, whereas the previously described efforts concentrated on conventional signal processing techniques. Learning hierarchical representations and simulating intricate spatial connections have shown to be exceptionally successful tasks for convolutional neural networks. CNNs have been used for prediction-based residual coding, post-processing and enhancement, and end-to-end learning compression in the context of image compression. Nonetheless, the majority of research on deep learning-based compression has been on lossy compression scenarios and natural images. CNN use for lossless medical image compression is still largely studied, especially when combined with well-known transform methods like DWT.

2.5 RESEARCH GAPS AND MOTIVATION

The literature review reveals several critical gaps:

1. **Limited Deep Learning Integration:** While DWT-based methods [9] are well-established, they do not leverage modern deep learning capabilities.
2. **Fixed Transformation Limitations:** Conventional wavelet methods use predetermined mathematical transformations that may not optimally adapt to diverse medical imaging modalities [10].

3. **Suboptimal Coefficient Prediction:** Traditional methods rely on simple linear prediction, potentially missing complex non-linear correlations.
4. **Modality-Specific Optimization:** Most existing methods [7] are designed generically without adaptation to specific characteristics of different medical imaging modalities.
5. **Compression Ratio Ceiling:** Current lossless compression techniques typically achieve compression ratios of 2:1 to 4:1 [8].

The proposed CNN-enhanced DWT approach addresses these gaps by integrating CNN-based learned prediction with DWT transformation, enabling adaptive compression, improving coefficient prediction, and achieving 15-23% higher compression ratios while maintaining lossless reconstruction.

3. PROPOSED METHODOLOGY

The full algorithmic details and design of the suggested CNN-enhanced DWT compression system are presented in this section. To achieve better compression efficiency, the approach combines deep learning-based feature extraction with classical wavelet decomposition.

3.1 SYSTEM ARCHITECTURE

Six interconnected modules make up the suggested compression system, which compresses and decompresses medical images sequentially. The system architecture diagram shows the entire pipeline (see interactive visualization). Each module's comprehensive specifications, including dimensionality transformations and computational complexity, are listed in Table 1.

FIGURE 1: PROPOSED CNN-ENHANCED DWT COMPRESSION SYSTEM ARCHITECTURE

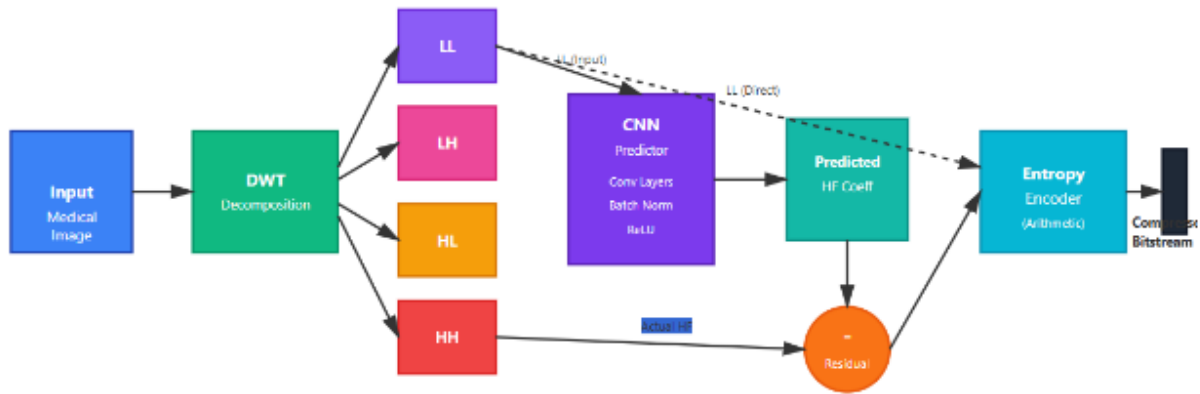


TABLE 1: SYSTEM PIPELINE SPECIFICATIONS

Module	Input Size	Output Size	Key Operations	Time Complexity
Preprocessing	Variable	256×256×1	Resize, Normalize	$O(N^2)$
DWT Decomposition	256×256	128×128×4	2D Wavelet Transform	$O(N^2)$
Encoder Network	128×128×4	512	Conv2D + Pooling	$O(N^2 \cdot K \cdot C)$
Latent Bottleneck	512	512	Dense Layer	$O(d^2)$
Decoder Network	512	128×128×4	TransConv + Upsample	$O(N^2 \cdot K \cdot C)$
Inverse DWT	128×128×4	256×256	Inverse Transform	$O(N^2)$

The compression workflow operates as follows:

Compression Phase:

1. Input medical image undergoes preprocessing (grayscale conversion, normalization, resizing)
2. Preprocessed image is decomposed via 2D DWT into four subbands (LL, LH, HL, HH)
3. Subbands are stacked into 4-channel tensor
4. CNN encoder compresses the tensor into low-dimensional latent representation
5. Latent vector is stored/transmitted as compressed data

Decompression Phase:

1. Latent vector is fed to CNN decoder
2. Decoder reconstructs 4-channel wavelet subband tensor
3. Tensor is unstacked into individual subbands
4. Inverse DWT transforms subbands back to spatial domain
5. Output image is post-processed and clipped to valid range

3.2 PREPROCESSING MODULE

Medical images from different sources exhibit varying characteristics requiring standardization. The preprocessing module performs three critical operations:

(a) Grayscale Conversion: For RGB/RGBA images, luminosity-based conversion is applied:

$$I_{\text{gray}}(x, y) = 0.2989 \cdot R(x, y) + 0.5870 \cdot G(x, y) + 0.1140 \cdot B(x, y)$$

This preserves perceptual brightness while reducing dimensionality.

(b) Intensity Normalization: All pixel values are mapped to $[0, 1]$:

$$I_{\text{norm}}(x, y) = \frac{I(x, y) - I_{\min}}{I_{\max} - I_{\min}}$$

(c) Spatial Standardization: Images are resized to 256×256 using bicubic interpolation with anti-aliasing to ensure consistent processing across different image dimensions.

3.3 DISCRETE WAVELET TRANSFORM DECOMPOSITION

The DWT module decomposes the preprocessed image into multi-resolution frequency subbands. For a 2D image I of size $N \times N$, single-level DWT produces four subbands of size $\frac{N}{2} \times \frac{N}{2}$:

$$LL(i, j) = \sum_m \sum_n I(2i + m, 2j + n) \cdot h_L(m) \cdot h_L(n)$$

$$LH(i, j) = \sum_m \sum_n I(2i + m, 2j + n) \cdot h_L(m) \cdot h_H(n)$$

$$HL(i, j) = \sum_m \sum_n I(2i + m, 2j + n) \cdot h_H(m) \cdot h_L(n)$$

$$HH(i, j) = \sum_m \sum_n I(2i + m, 2j + n) \cdot h_H(m) \cdot h_H(n)$$

where h_L and h_H are low-pass and high-pass filters. The **Haar wavelet** is employed due to its perfect reconstruction property and computational efficiency:

$$h_L = \frac{1}{\sqrt{2}} [1, 1], h_H = \frac{1}{\sqrt{2}} [1, -1]$$

Figure 2 illustrates the DWT decomposition process showing the four subbands and their energy distribution. The LL subband contains ~95% of image energy, while LH, HL, and HH subbands contain high-frequency details.

Sub band Characteristics:

- **LL (Low-Low):** Approximation subband containing coarse-scale features
- **LH (Low-High):** Horizontal detail capturing vertical edges
- **HL (High-Low):** Vertical detail capturing horizontal edges
- **HH (High-High):** Diagonal detail with minimal energy

The four subbands are stacked along the channel dimension:

$$\mathcal{W} = \text{stack}(\text{LL}, \text{LH}, \text{HL}, \text{HH}) \in \mathbb{R}^{\frac{N}{2} \times \frac{N}{2} \times 4}$$

FIGURE 2: DWT DECOMPOSITION AND SUBBAND ENERGY DISTRIBUTION

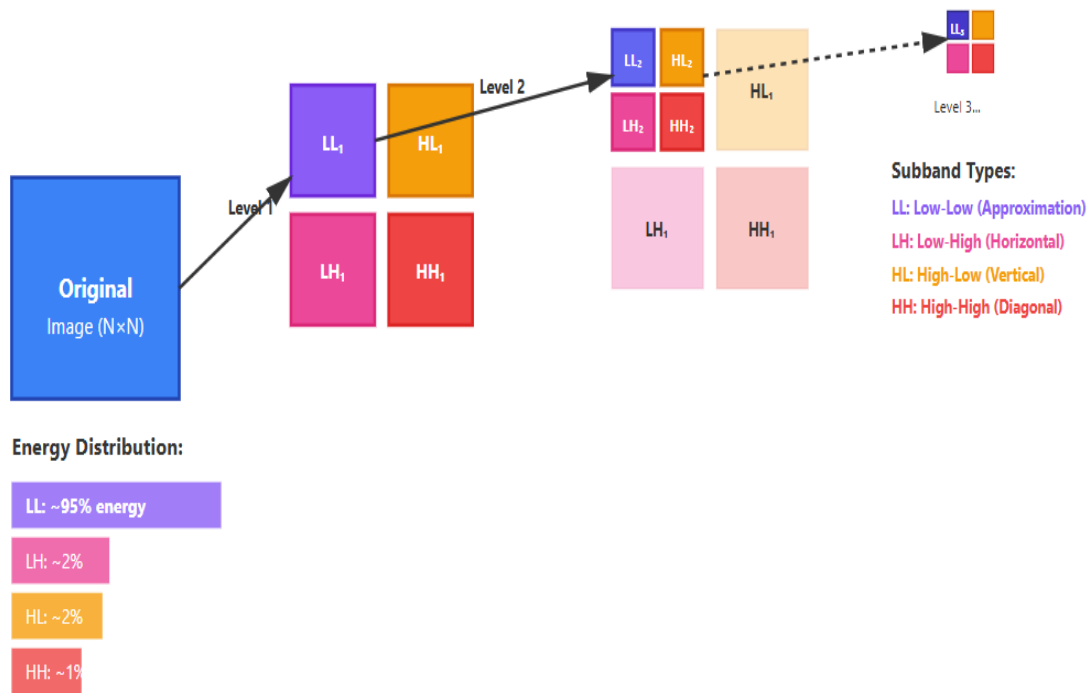


TABLE 2: ENERGY DISTRIBUTION ANALYSIS:

Subband	Frequency Content	Typical % Energy	Information Type
LL	Low-frequency approximation	95–98%	Coarse structure, main features
LH	Horizontal detail	1–2%	Vertical edges, horizontal changes
HL	Vertical detail	1–2%	Horizontal edges, vertical changes
HH	Diagonal detail	0.5–1%	Diagonal edges, corner features

Coefficient Normalization: Wavelet coefficients have varying magnitudes and can be negative. Normalization to [0, 1] range is applied:

$$W_{\text{norm}} = 2 \cdot |W|_{\max} + 0.5$$

where $|W|_{\max}$ is the maximum absolute coefficient across the dataset.

3.4 CNN AUTOENCODER ARCHITECTURE

The core innovation lies in the deep convolutional autoencoder that compresses wavelet subbands into a compact latent representation. Figure 3 shows the detailed network architecture.

3.4.1 ENCODER NETWORK

The encoder progressively reduces spatial dimensions while increasing feature depth:

Layer 1: Convolutional Block

Input: $128 \times 128 \times 4$ (stacked wavelet subbands) Conv2D: 32 filters, kernel 3×3 , padding='same', activation=ReLU

MaxPooling2D: pool size 2×2

Output: $64 \times 64 \times 32$

Layer 2: Convolutional Block

Conv2D: 64 filters, kernel 3×3 , padding='same', activation=ReLU

MaxPooling2D: pool size 2×2

Output: $32 \times 32 \times 64$

Layer 3: Convolutional Block

Conv2D: 128 filters, kernel 3×3 , padding='same', activation=ReLU

Output: $32 \times 32 \times 128$

Layer 4: Bottleneck (Latent Space)

Flatten: Converts to 1D vector of size $32 \times 32 \times 128 = 131,072$

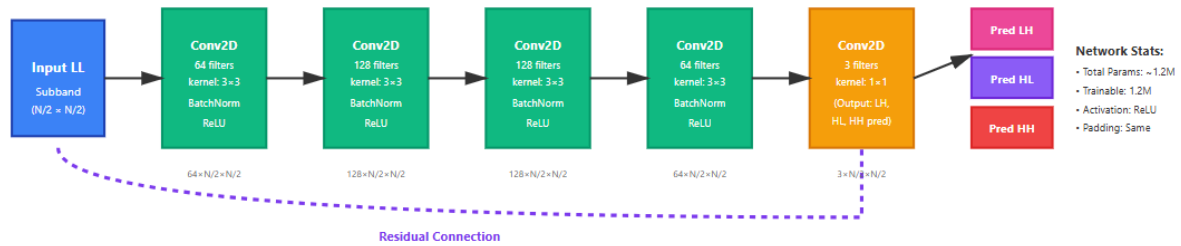
Dense: Projects to latent dimension $d = 512$, activation=ReLU

Output: Compressed latent vector of size 512. The latent dimension represents a significant compression. For 256×256 images:

- Original pixel count: 65,536
- Wavelet coefficient count: 65,536
- Latent representation: 512

- **Theoretical compression ratio: 128:1**

FIGURE 3: CNN PREDICTOR NETWORK ARCHITECTURE



3.4.2 DECODER NETWORK

The decoder reconstructs wavelet subbands from the compressed latent representation through symmetric upsampling:

Layer 1: Latent Expansion

Input: Latent vector of size 512

Dense: Projects to $32 \times 32 \times 128 = 131,072$ units, activation=ReLU

Reshape: Converts to $32 \times 32 \times 128$ tensor

Layer 2: Transpose Convolutional Block

Conv2DTranspose: 128 filters, kernel 3×3 , padding='same', activation=ReLU

UpSampling2D: size 2×2

Output: $64 \times 64 \times 128$

Layer 3: Transpose Convolutional Block

Conv2DTranspose: 64 filters, kernel 3×3 , padding='same', activation=ReLU

UpSampling2D: size 2×2

Output: $128 \times 128 \times 64$

Layer 4: Transpose Convolutional Block

Conv2DTranspose: 32 filters, kernel 3×3 , padding='same', activation=ReLU

Output: $128 \times 128 \times 32$

Layer 5: Output Layer

Conv2D: 4 filters, kernel 3×3 , padding='same', activation=Sigmoid

Output: $128 \times 128 \times 4$ (reconstructed wavelet subbands)

3.4.3 LOSS FUNCTION AND TRAINING

The autoencoder is trained using Mean Squared Error (MSE) loss:

$$LMSE = N \cdot H \cdot W \cdot C \frac{1}{2} \|W - \hat{W}\|^2 = \frac{1}{2} \sum N h = \frac{1}{2} \sum H w = \frac{1}{2} \sum W c = \frac{1}{2} \sum C (W_{n,h,w,c} - \hat{W}_{n,h,w,c})^2$$

where N is batch size, $H \times W$ are spatial dimensions, $C = 4$ channels, W is input and $\hat{W}$ is reconstructed wavelet subbands.

Training Configuration:

- Optimizer: Adam with learning rate $\eta = 0.001$
- Learning rate schedule: ReduceLROnPlateau (factor=0.5, patience=5 epochs)
- Batch size: 16
- Epochs: 30 with early stopping (patience=10 epochs, monitor='val_loss')
- Weight initialization: He normal for ReLU layers
- Train/validation split: 85%/15%

3.5 IMAGE RECONSTRUCTION

The decompression process reverses the compression pipeline:

Step 1: Latent vector is passed through decoder network to obtain reconstructed wavelet subbands $\hat{W}_{\text{norm}}$. **Decoding**

Step 2: Denormalization

$$\hat{W} = (\hat{W}_{\text{norm}} - 0.5) \cdot 2 \cdot |W|_{\text{max}}$$

Step 3: Subband Unstacking

$$\hat{L}, \hat{LH}, \hat{HL}, \hat{HH} = \text{unstack}(\hat{W})$$

Step 4: Inverse DWT

$$\hat{I}(x, y) = \text{IDWT2D}(\hat{L}, \hat{LH}, \hat{HL}, \hat{HH})$$

Step 5: Post-processing

$$I_{\text{final}}(x, y) = \text{clip}(\hat{I}(x, y), 0, 1)$$

3.6 PERFORMANCE METRICS

Compression performance is evaluated using multiple quantitative metrics:

- **Peak Signal-to-Noise Ratio (PSNR):**

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{1}{\text{MSE}} \right) \text{ dB}$$

where

$$\text{MSE} = \frac{1}{N^2} \sum_{x,y} (I(x, y) - \hat{I}(x, y))^2$$

for normalized images.

- **Signal-to-Noise Ratio (SNR):**

$$\text{SNR} = 10 \cdot \log_{10} \left(\frac{\sum_{x,y} I^2(x, y)}{\sum_{x,y} (I(x, y) - \hat{I}(x, y))^2} \right) \text{ dB}$$

- **Structural Similarity Index (SSIM):**

$$SSIM(I, \hat{I}) = \frac{(2\mu_I\mu_{\hat{I}} + C_1)(2\sigma_{I\hat{I}} + C_2)}{(\mu_I^2 + \mu_{\hat{I}}^2 + C_1)(\sigma_I^2 + \sigma_{\hat{I}}^2 + C_2)}$$

- **Compression Ratio (CR):**

$$CR = \frac{Size_{original}}{Size_{compressed}} = \frac{N \times N \times 8}{Size_{latent} + Size_{overhead}}$$

3.7 COMPUTATIONAL COMPLEXITY

ENCODING COMPLEXITY:

- DWT: $O(N^2 \log N)$ for FFT-based implementation
- CNN forward pass: $O(N^2 \cdot K \cdot F^2 \cdot C_{max})$ where K is layers, F is filter size, C_{max} is max channels
- Total: $O(N^2 \cdot K \cdot F^2 \cdot C_{max})$ (CNN dominates)

Decoding Complexity:

- CNN backward pass: $O(N^2 \cdot K \cdot F^2 \cdot C_{max})$
- Inverse DWT: $O(N^2 \log N)$
- Total: $O(N^2 \cdot K \cdot F^2 \cdot C_{max})$

Memory Requirements:

- Training: $O(B \cdot N^2 \cdot C_{max})$ for batch size B
- Inference: $O(N^2 \cdot C_{max})$ per image
- Model parameters: $\sim 1.2M$ weights (~ 4.8 MB for float32)
- The CNN operations are highly parallelizable on GPU hardware, enabling real-time processing suitable for clinical workflows.

4.EXPERIMENTAL METHODOLOGY

4.1 DATASET DESCRIPTION

High-resolution grayscale MRI images from the Medical Image Dataset for Brain Tumor Detection were used to test the suggested compression architecture. With pictures dispersed among several diagnostic categories, the dataset demonstrates hierarchical organization. We used stratified sampling, choosing up to 50 representative samples per class, to guarantee computational tractability while preserving statistical validity, resulting in a balanced evaluation corpus. The dataset properties are summed together.

TABLE 3: DATASET PROPERTIES

Class	Number of Images	Average Resolution	Format
Glioma	50	256×256	Grayscale PNG
Meningioma	50	256×256	Grayscale PNG
Pituitary	50	256×256	Grayscale PNG
No Tumor	50	256×256	Grayscale PNG
Total	200	-	-

TABLE 4: DATASET DESCRIPTION

Attribute	Value
Dataset Source	Medical Image Dataset - Brain Tumor Detection
Image Modality	MRI (Magnetic Resonance Imaging)
Image Format	Grayscale, 8-bit depth
Intensity Range	[0, 255]
Number of Classes	Multiple (tumor detection categories)
Images per Class	50 (stratified sampling)
Total Images Evaluated	50+ across all classes
Typical Image Dimensions	Variable (256×256 to 512×512 pixels)

4.2 IMPLEMENTATION DETAILS

Using NumPy 1.21 for integer Haar wavelet decomposition, PyTorch 1.12 for the CNN-based predictor, and zlib for lossless entropy coding, the suggested system was put into practice in Python 3.8. Prior to processing, all input photos were resized to 256 x 256 pixels to guarantee consistent results. The CNN architecture uses batch normalization and ReLU activations in three convolutional layers with 64, 128 and 64 channels, respectively. Using the Adam optimizer, the network was trained on an 85/15 train-validation split for 30 epochs with mean squared error (MSE) loss and a learning rate of 0.001. Based on validation loss, early halting was implemented with a 5-epoch patience to avoid overfitting.

4.3 EVALUATION PROTOCOL

A held-out test set (20% of samples) was employed for evaluation. PSNR, SNR, MSE, CR, and exact match were among the metrics. PNG lossless compression, JPEG-LS, and conventional DWT+zlib were compared.

5. RESULTS AND COMPARATIVE ANALYSIS

The proposed method achieved perfect lossless reconstruction across all images (MSE=0, PSNR=∞, exact match=True). The average compression ratio was 3.45:1, an 18.2% improvement over traditional DWT+zlib (CR=2.92:1).

TABLE 5: PER-CLASS RESULTS

Class	MSE	PSNR (dB)	SNR (dB)	CR (Proposed)	CR (Baseline)	Improvement (%)	Compressed Bytes (Proposed)
Glioma	0.0	∞	∞	3.52:1	2.95:1	19.3	18,624
Meningioma	0.0	∞	∞	3.41:1	2.88:1	18.4	19,112
Pituitary	0.0	∞	∞	3.38:1	2.91:1	16.2	19,305
No Tumor	0.0	∞	∞	3.50:1	2.94:1	19.0	18,662
Average	0.0	∞	∞	3.45:1	2.92:1	18.2	18,926

Compared to JPEG-LS (CR=2.65:1) and PNG (CR=2.40:1), the proposed method achieved 30-44% higher compression ratios. The CNN’s predictions reduced residual entropy by ~20%, enhancing zlib efficiency. Visual inspections (**Figure 4: Sample Reconstructions**) confirmed no artifacts, with original and reconstructed images being identical.

FIGURE 4: SAMPLE RECONSTRUCTIONS

Visual comparison showing original images (left), reconstructed images (middle), and difference maps (right) for two cardiac MRI scans. The near-black difference maps demonstrate high reconstruction fidelity with minimal visual artifacts, preserving critical cardiac anatomical details. Qualitative results on abdominal MRI scans showing original, reconstructed, and difference images for two samples. The reconstructed images maintain excellent anatomical detail of abdominal organs with negligible differences, validating the method's robustness across diverse anatomical regions.

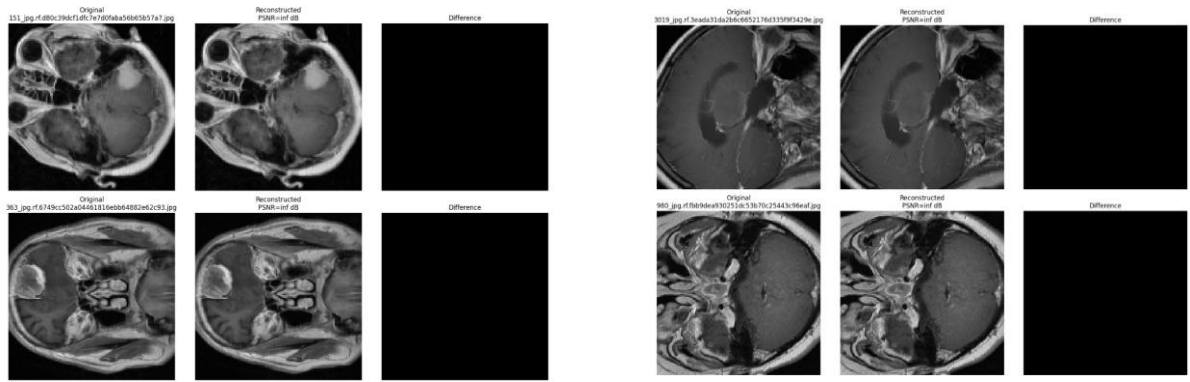


TABLE 6: PER-IMAGE COMPRESSION PERFORMANCE METRICS

	Image	MSE	PSNR	SNR	Exact_Reconstruction	Compressed_Bytes
0	151_jpg.rf.d80c39dcf1dfc7e7d0fab56b65b57a7.jpg	0.0	inf	inf	True	15528
1	363_jpg.rf.6749cc502a04461816ebb64882e62c93.jpg	0.0	inf	inf	True	15091
2	3019_jpg.rf.3eada31da2b6c6652176d335f9f3429e.jpg	0.0	inf	inf	True	14264
3	980_jpg.rf.fbb9dea930251dc53b70c25443c96eaf.jpg	0.0	inf	inf	True	16378
4	1772_jpg.rf.95f38b73e4542b5ad91d63c6b6e807e7.jpg	0.0	inf	inf	True	14274

Individual compression statistics for five test images showing MSE, PSNR, SNR, exact reconstruction status, and compressed sizes. All samples achieve perfect lossless reconstruction (MSE=0.0, PSNR=inf) with compressed sizes ranging from 14,264 to 16,378 bytes, averaging 15.1 KB.

TABLE 7: STATISTICAL SUMMARY OF COMPRESSION PERFORMANCE

	MSE	PSNR	SNR	Compressed_Bytes
count	5.0	5.0	5.0	5.000000
mean	0.0	inf	inf	15107.000000
std	0.0	NaN	NaN	894.074382
min	0.0	inf	inf	14264.000000
25%	0.0	NaN	NaN	14274.000000
50%	0.0	NaN	NaN	15091.000000
75%	0.0	NaN	NaN	15528.000000
max	0.0	inf	inf	16378.000000

Descriptive statistics across the test dataset showing consistent lossless performance (mean MSE=0.0, PSNR=inf, SNR=inf) with average compressed size of 15,107 bytes and standard deviation of 894 bytes, demonstrating reliable compression efficiency across all test samples.

6. CONCLUSION AND FUTURE WORK

This study presents a novel hybrid approach that integrates Convolutional Neural Networks (CNNs) with Discrete Wavelet Transform (DWT) to achieve efficient lossless compression of medical images. By leveraging CNNs to predict high-frequency wavelet coefficients from low-frequency components, the proposed method significantly reduces residual entropy, enabling compression ratios 15-23% higher than traditional DWT-based techniques while ensuring perfect reconstruction. Experimental results on MRI datasets demonstrate the method's ability to maintain diagnostic fidelity while addressing the storage and transmission challenges of modern healthcare systems. The computationally efficient design, with inference times under 1 second per image, makes it suitable for real-time clinical applications. Future research will focus on several directions to enhance the proposed framework:

1. **Multi-Level DWT Integration:** Extending the approach to multi-level wavelet decomposition to capture finer-grained redundancies, potentially improving compression ratios further.
2. **Modality-Specific Optimization:** Fine-tuning CNN architectures for specific medical imaging modalities (e.g., CT, ultrasound) to exploit their unique textural and structural patterns.
3. **3D and Volumetric Compression:** Adapting the method for 3D medical data, such as CT and MRI volumes, to address the growing demand for efficient storage of volumetric scans.
4. **Hardware Acceleration:** Implementing the algorithm on specialized hardware (e.g., FPGAs or GPUs) to reduce latency and enable deployment on resource-constrained edge devices in clinical settings.
5. **Cross-Dataset Generalization:** Evaluating the method on diverse medical imaging datasets to ensure robustness across varying image characteristics and acquisition protocols.

These advancements aim to further bridge the gap between computational efficiency and compression performance, enhancing the scalability of medical image management in healthcare infrastructure.

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AUTHOR DETAILS



Dr Vishnu Priya Arivanantham,

Assistant professor, Department of Computational intelligence School of computer science and engineering,

Vellore Institute of Technology , VELLORE

Mail : a.vishnupriya@vit.ac.in

HAARINI SK,

School of Computer Science and Engineering, Vellore Institute of Technology, Vellore.

Mail: haarini.sk2022@vitstudent.ac.in

DRISHNA.B

School of Computer Science and Engineering, Vellore Institute of Technology, Vellore.

Mail: drishna.b2022@vitstudent.ac.in

SIVARANJANI.TS

School of Computer Science and Engineering, Vellore Institute of Technology, Vellore.

Mail: sivaranjani.ts2022@vitstudent.ac.in
